UNIVERZITET U NOVOM SADU
FAKULTET TEHNIČKIH NAUKA
KATEDRA ZA AUTOMATIKU I UPRAVLJANJE SISTEMIMA

Linearan matematički model Primena Laplasove transformacije (deo A)

Modeliranje i simulacija sistema

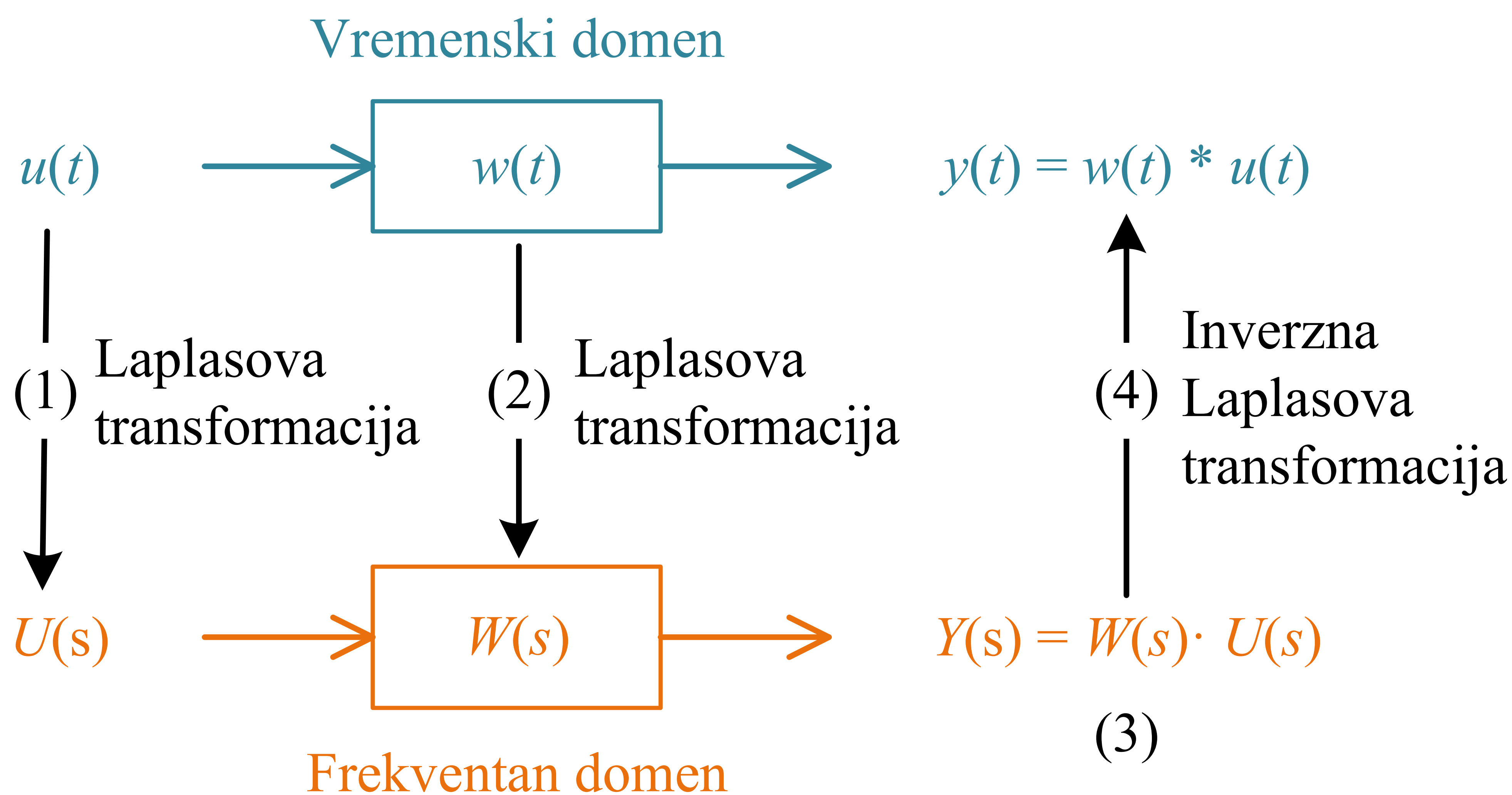
Upravljanje, modelovanje i simulacija sistema

Pierre-Simon Laplace

Pierre-Simon Laplace (1749–1827). Posthumous portrait by Jean-Baptiste Paulin Guérin, 1838.

Born	(1749-03-23)23 March 1749 Beaumont-en-Auge, Normandy, Kingdom of France
Died	5 March 1827(1827-03-05) (aged 77) Paris, Bourbon France
Nationality	French
Alma mater	University of Caen
Known for	Laplace transform …
	Scientific career
Fields	Astronomer and mathematician
Institutions	École Militaire (1769–1776)
Academic advisors	Jean d'Alembert Christophe Gadbled Pierre Le Canu
Doctoral students	Siméon Denis Poisson

Proces primene Laplasove transformacije



Definicija Laplasove transformacije

- Definicija

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0}^{\propto} f(t)e^{-s}\, dt$$

  - $\mathcal{L}$ – operator Laplasove transformacije (kraće Laplasov operator),
  - s – kompleksna učestanost ili kompleksna promenljiva Laplasove transformacije,
  - $f(t)$ – vremenski zavisan signal - **original**,
  - $F(s)$ – **kompleksan lik** signala $f(t)$.

- Inverzna Laplasova transformacija

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{\gamma - j\omega}^{\gamma + j\omega} F(s)e^{st} ds, \qquad t > 0$$

Tablica Laplasove transformacije

Br.	Original $f(t), (t > 0)$	Kompleksan lik $F(s)$
1	$\delta(t)$	1
2	$t^n, \qquad n = 0,1,2, \ldots$	$\frac{n!}{s^{n+1}}$
3	e^{-at}	$\frac{1}{s+a}$
4	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
5	$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
6	$\mathrm{e}^{-\alpha \mathrm{t}}\cos(\omega t)$	$\frac{s+\alpha}{(s+\alpha)^2+\omega^2}$
7	$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
8	$\mathrm{e}^{-\alpha \mathrm{t}}\sin(\omega t)$	$\frac{\omega}{(s+\alpha)^2+\omega^2}$

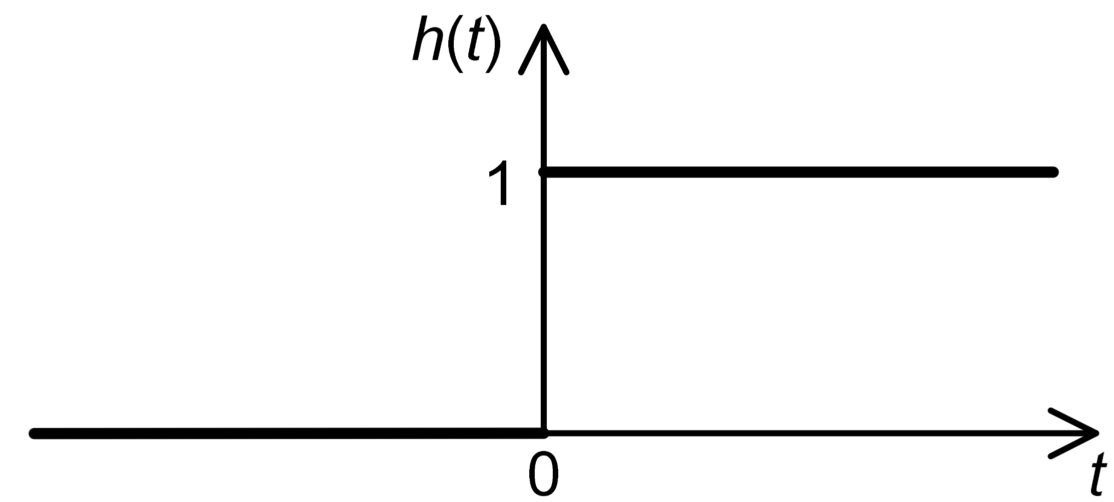
Osobine Laplasove transformacije

Br.	Naziv osobine	Izraz
1	Linearnost	$\mathcal{L}\{a_1 f_1(t) + a_2 f_2(t)\} = a_1 F_1(s) + a_2 F_2(s)$
2	Čisto vremensko kašnjenje	$\mathcal{L}\{f(t-\tau)\} = e^{-s\tau} F(s)$
3	Pomeranje kompleksnog lika	$\mathcal{L}\{e^{-at} f(t)\} = F(s+a)$
4	Konvolucija originala	$\mathcal{L}\{f_1(t) * f_2(t)\} = F_1(s) F_2(s)$
5	Izvod originala	$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0)$ $\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s) - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)$
6	Integral originala	$\mathcal{L}\left\{\int_0^t f(t)dt\right\} = \frac{F(s)}{s}$ $\mathcal{L}\left\{\iint_0^t \dots \int_0^t f(t)dt^n\right\} = \frac{F(s)}{s^n}$
7	Izvod kompleksnog lika	$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}$
8	Promena vremenske skale	$\mathcal{L}\left\{f\left(\frac{t}{a}\right)\right\} = aF(as)$
9	Granične vrednosti	$\lim_{t \to 0} f(t) = \lim_{s \to \propto} sF(s)$ $\lim_{t \to \propto} f(t) = \lim_{s \to 0} sF(s)$

Hevisajdov signal

- Poznat kao jedinični signal

$$h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$



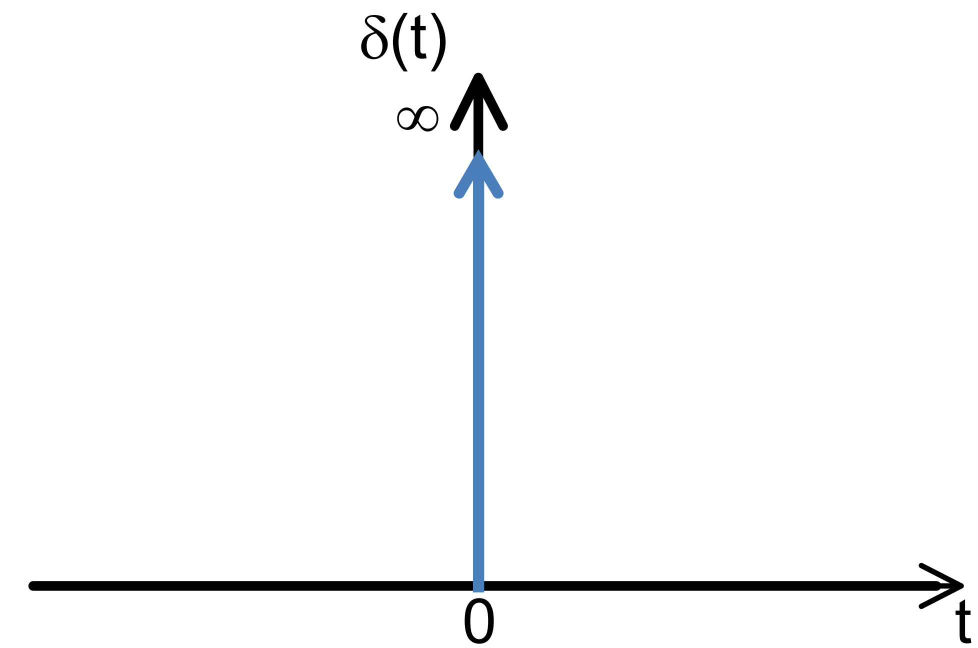
$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}dt$$

$$H(s) = \mathcal{L}\{h(t)\} = \int_0^\infty 1 \cdot e^{-st}dt = -\frac{e^{-st}}{s}\bigg|_0^\infty = \frac{1}{s}$$

Dirakov impuls

$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$$

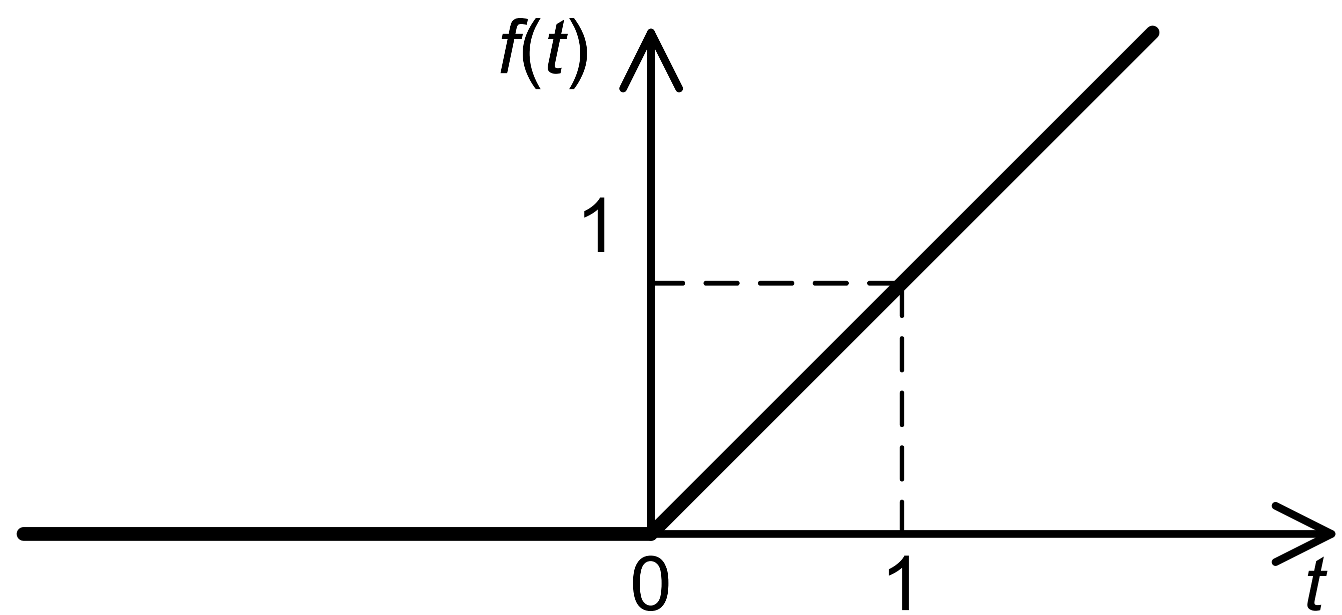
$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$



$$D(s) = \mathcal{L}\{\delta(t)\} = \int_{0-}^{\infty} \delta(t)e^{-st}dt = \int_{0-}^{\infty} \delta(t)e^{0}dt = \int_{-\infty}^{\infty} \delta(t)dt = 1$$

$$D(s) = \mathcal{L}\left\{\frac{dh(t)}{dt}\right\} = s\mathcal{L}\{h(t)\} - h(0_{-}) = s \cdot \frac{1}{s} - 0 = 1$$

Jedinični nagibni signal



$$f(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases}$$

$$F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{t \cdot h(t)\} = \int_0^\infty t \cdot e^{-st} dt = -\frac{t \cdot e^{-s}}{s}\bigg|_0^\infty + \frac{1}{s}\int_0^\infty e^{-st} dt = \frac{1}{s^2}$$

$$u = t, dv = e^{-s}\ dt$$

Transformacije poznatih signala

- $f(t) = \delta(t),$ $\qquad F(s) = \int_{0-}^{\infty} \delta(t) e^{-st} dt = 1,\ \int_{-\infty}^{\infty} \delta(t) dt = 1$
- $f(t) = h(t),$ $\qquad F(s) = \int_{0}^{\infty} e^{-st} dt = -\frac{1}{s} e^{-st} \big|_{0}^{\infty} = \frac{1}{s}$
- $f(t) = t,$ $\qquad F(s) = \int_{0}^{\infty} t e^{-st} dt = -t \left.\frac{e^{-st}}{s}\right|_{0}^{\infty} + \frac{1}{s} \int_{0}^{\infty} e^{-st} dt = \frac{1}{s^2}$
- $f(t) = t^2,$ $\qquad F(s) = \int_{0}^{\infty} t^2 e^{-st} dt = -t^2 \left.\frac{e^{-s}}{s}\right|_{0}^{\infty} + \frac{2}{s} \int_{0}^{\infty} t e^{-st} dt = \frac{2}{s^3}$

  $u = t^2, dv = e^{-st} dt$
- $f(t) = e^{at},$ $\qquad F(s) = \int_{0}^{\infty} e^{at} e^{-st} dt = \int_{0}^{\infty} e^{-(s-a)t} dt = -\frac{1}{s-a} e^{-(s-a)t} \big|_{0}^{\infty} = \frac{1}{s-a}$
- $f(t) = \cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2},$ $\qquad F(s) = \int_{0}^{\infty} \frac{e^{j\omega t} + e^{-j\omega t}}{2} e^{-st} dt = \frac{1}{2} \left(\frac{1}{s - j\omega} + \frac{1}{s + j\omega} \right) = \frac{s}{s^2 + \omega^2}$
- $f(t) = \sin \omega t = \frac{e^{j\omega t} - e^{-j\omega}}{2j},$ $\qquad F(s) = \frac{1}{2j} \left(\frac{1}{s - j\omega} - \frac{1}{s + j\omega} \right) = \frac{\omega}{s^2 + \omega^2}$

Osobina: Linearnost

$$\mathcal{L}\{a_1 f_1(t) + a_2 f_2(t)\} = a_1 F_1(s) + a_2 F_2(s)$$

$$\mathcal{L}\{a_1 f_1(t) + a_2 f_2(t)\} =$$

$$= \int_0^{\infty} \big(a_1 f_1(t) + a_2 f_2(t)\big) e^{-st} dt =$$

$$= a_1 \int_0^{\propto} f_1(t) e^{-st} dt + a_2 \int_0^{\propto} f_2(t) e^{-st} dt =$$

$$= a_1 \mathcal{L}\{f_1(t)\} + a_2 \mathcal{L}\{f_2(t)\}$$

$$= a_1 F_1(s) + a_2 F_2(s)$$

Osobina: Čisto vremensko kašnjenje

$$\mathcal{L}\{f(t-\tau)\} = e^{-s\tau}F(s)$$

$\mathcal{L}\{f(t-\tau)\} =$

$= \int_0^\infty f(t-\tau)e^{-st}dt =$ Smena: $t-\tau = v$

$= e^{-s\tau}\int_{-\tau}^\infty f(v)e^{-sv}dv =$ kauzalan signal: $f(v) \equiv 0, v < 0$

$= e^{-s\tau}\int_0^\infty f(v)e^{-sv}dv =$

$= e^{-s\tau}F(s)$

Osobina: Pomeranje kompleksnog lika

$$\mathcal{L}\{e^{-at}f(t)\} = F(s+a)$$

$$\mathcal{L}\{e^{-at}f(t)\} = \int_0^\infty f(t)e^{-(s+a)t}dt = F(s+a)$$

Posledica: $\mathcal{L}$ signala $e^{-at}, e^{-at}\sin\omega t\,, e^{-at}\cos\omega t$ se svodi na zamenu s sa $s+a$ u kompleksnim likovima.

Osobina: Konvolucija originala

$$\mathcal{L}\{f_1(t) * f_2(t)\} = F_1(s)F_2(s)$$

- Konvolucija se definiše kao: $f_1(t) * f_2(t) = \int_0^t f_1(\tau)f_2(t-\tau)d\tau = \int_0^t f_1(t-\tau)f_2(\tau)d\tau$

$\mathcal{L}\{f_1(t) * f_2(t)\} =$

$$= \int_0^\infty e^{-st} \int_0^t f_1(t-\tau)f_2(\tau)d\tau\, dt =$$

$$= \int_0^\infty e^{-st} \int_0^\infty f_1(t-\tau)f_2(\tau)h(t-\tau)d\tau\, dt =$$

gornja granica integrala po τ se može proširiti sa t na ∞, jer je $f_1(t-\tau) \equiv 0$, za $t-\tau < 0$ tj. $t < \tau$ čime je podintegralna funkcija 0 za $t < \tau$.
Sada se zamene mesta integralima ...

$$= \int_0^\infty f_2(\tau) \int_0^\infty f_1(t-\tau)e^{-st}h(t-\tau)dt\, d\tau = \int_0^\infty f_2(\tau) \int_{-\tau}^\infty f_1(v)e^{-s(v+\tau)}h(v)dv\, d\tau$$

Uvodi se smena $v = t - \tau$,

$$= \int_0^\infty f_2(\tau)e^{-s\tau}\left(\int_{-\tau}^\infty f_1(v)e^{-sv}h(v)dv\right)d\tau =$$

$$= \int_0^\infty f_1(v)e^{-sv}dv \int_0^\infty f_2(\tau)e^{-s\tau}d\tau = F_1(s)F_2(s)$$

Donja granica dv integrala se može pomeriti u 0 jer je $h(v)$ nula za $t < 0$.

Osobina: Izvod originala

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0)$$

Parcijalni izvod: $\frac{d}{dt}\left(f(t)e^{-st}\right) = \frac{df(t)}{dt}e^{-st} - f(t)se^{-st}$, tj.

$$\frac{df(t)}{dt}e^{-st} = \frac{d}{dt}\left(f(t)e^{-st}\right) + f(t)se^{-st}$$

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} =$$

$$\int_0^\infty \left(\frac{df(t)}{dt}\right) e^{-st} dt =$$

$$\int_0^\infty \frac{d}{dt}\left(f(t)e^{-st}\right) dt + s\int_0^\infty f(t)e^{-st} dt =$$

$$f(t)e^{-st}\Big|_0^\infty + sF(s) =$$

$$0 - f(0) + sF(s)$$

Viši izvodi:

$$\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s) - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)$$

npr.

$$\mathcal{L}\left\{\frac{d^2 f(t)}{dt^2}\right\} = s^2 F(s) - sf(0) - \frac{df(0)}{dt}$$

Osobina: Integral originala

$$\mathcal{L}\left\{\int_0^t f(t)dt\right\} = \frac{F(s)}{s}$$

$F(s) = \int_0^{\infty} f(t)e^{-st}dt =$ Primeni se parcijalna integracija: $u = e^{-st}, dv = f(t)dt$

$= \left(e^{-st}\int_0^t f(\tau)d\tau\right)\Big|_0^{\infty} + s\int_0^{\propto} e^{-st}\int_0^t f(\tau)d\tau\, dt =$

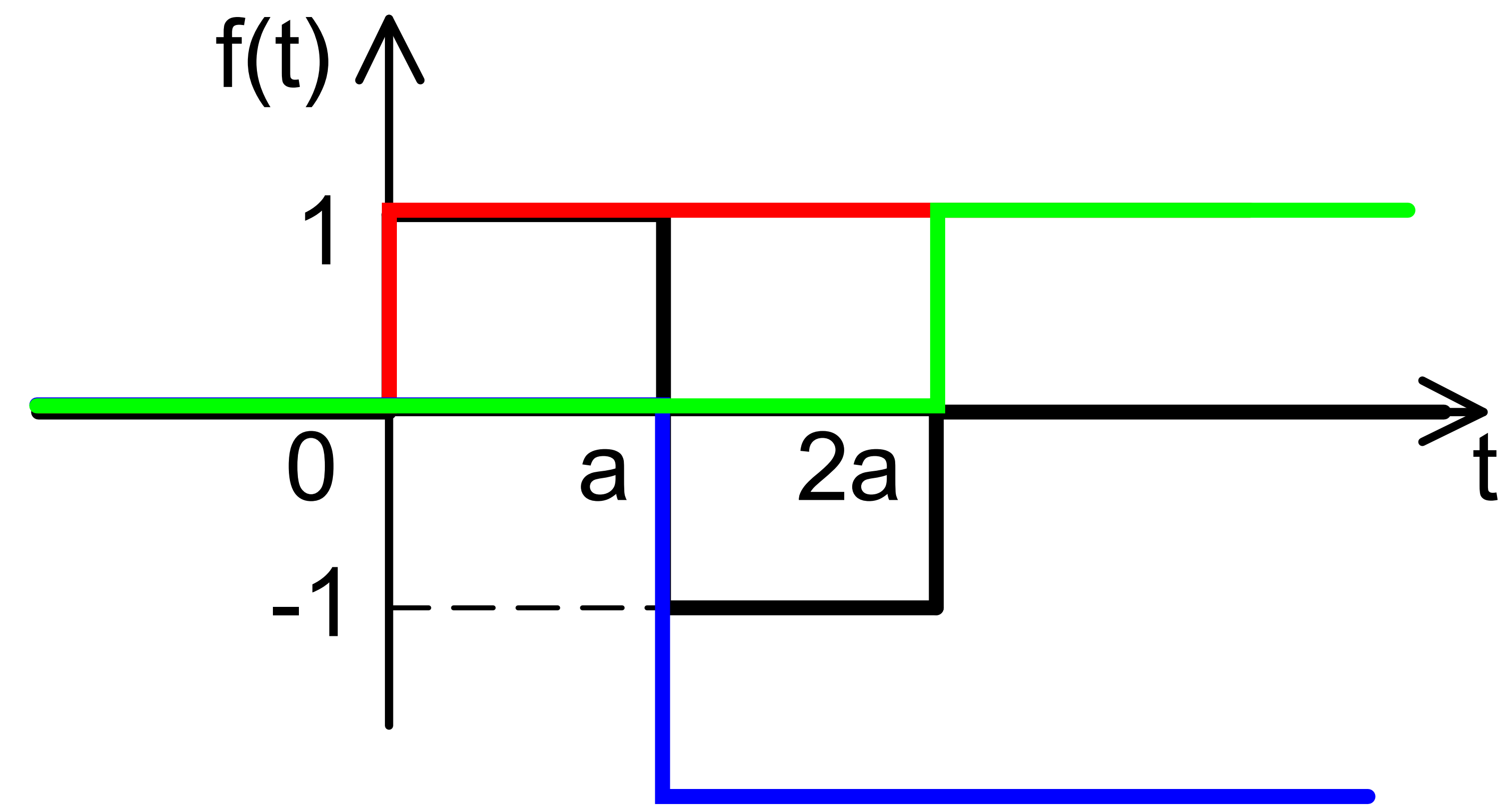
$= s\mathcal{L}\left\{\int_0^t f(t)dt\right\}$ i podeli se sa s

Višestruki integral:

$$\mathcal{L}\left\{\iint_0^t \dots \int_0^t f(t)dt^n\right\} = \frac{F(s)}{s^n}$$

Primer – Primena $\mathcal{L}$ na složen signal 1

$$f(t) = \begin{cases} 0, & t < 0 \\ 1, & 0 \le t < a \\ -1, & a \le t < 2a \\ 0, & t \ge a \end{cases}$$

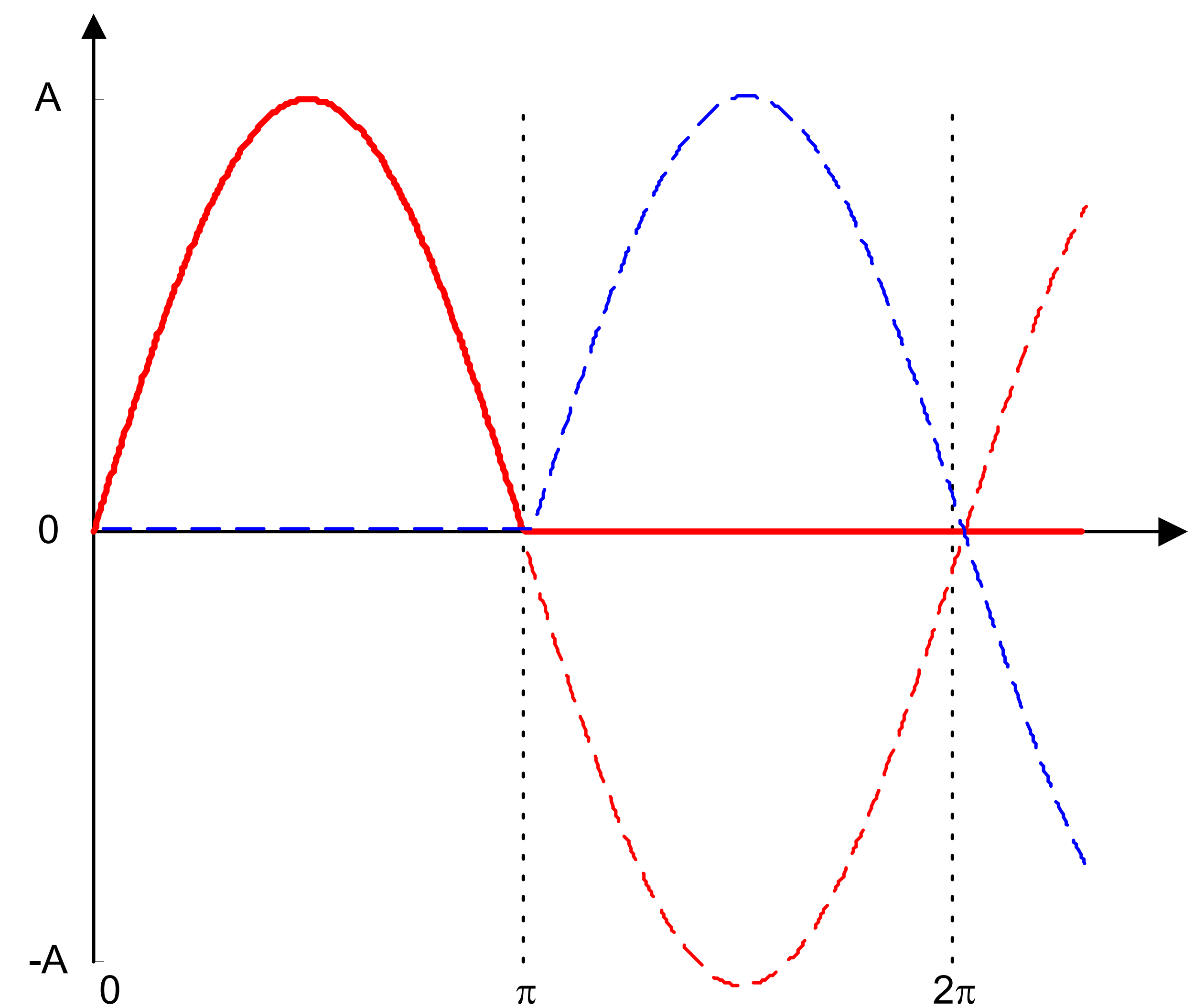


$$f(t) = h(t) - 2h(t-a) + h(t-2a)$$

$$F(s) = \mathcal{L}\{f(t)\} = \frac{1}{s}(1 - 2e^{-as} + e^{-2as})$$

Primer – Primena $\mathcal{L}$ na složen signal 2

$$f(t) = \begin{cases} 0, & t < 0 \\ A\sin t, & 0 \le t < \pi \\ 0, & t \ge \pi \end{cases}$$



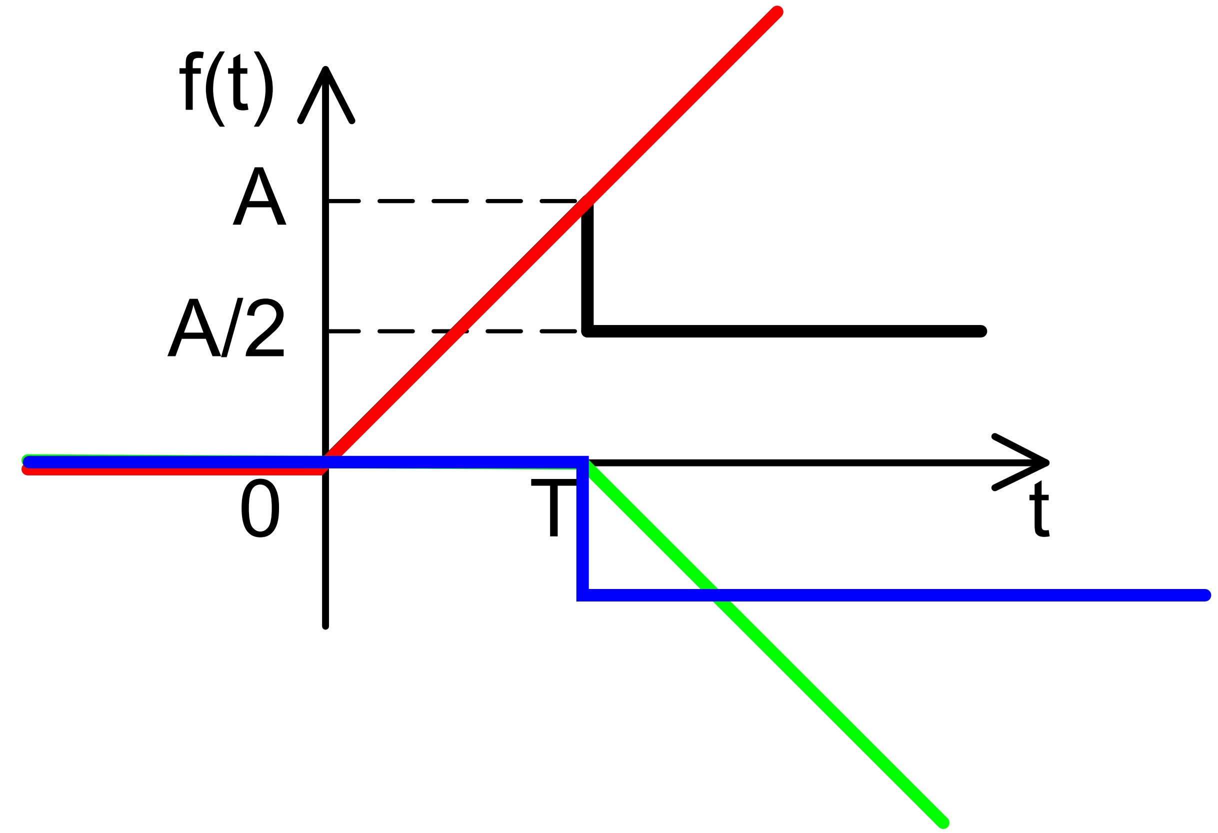
$$f(t) = A\sin t \cdot h(t) + A\sin(t-\pi)\cdot h(t-\pi)$$

$$F(s) = \frac{A}{s^2+1} + \frac{Ae^{-s\pi}}{s^2+1} = \frac{A}{s^2+1}(1+e^{-s\pi})$$

$$\begin{aligned} f(t) &= A\sin t \cdot [h(t) - h(t-\pi)] \\ &= A\sin t \cdot h(t) - A\sin t \cdot h(t-\pi) \\ &= A\sin t \cdot h(t) - A\sin(t-\pi+\pi)\cdot h(t-\pi) \\ &= A\sin t \cdot h(t) - A\sin(t-\pi)\cos\pi \cdot h(t-\pi) - A\cos(t-\pi)\sin\pi \cdot h(t-\pi) \\ &= A\sin t \cdot h(t) + A\sin(t-\pi)\cdot h(t-\pi) \end{aligned}$$

Primer – Primena $\mathcal{L}$ na složen signal 3

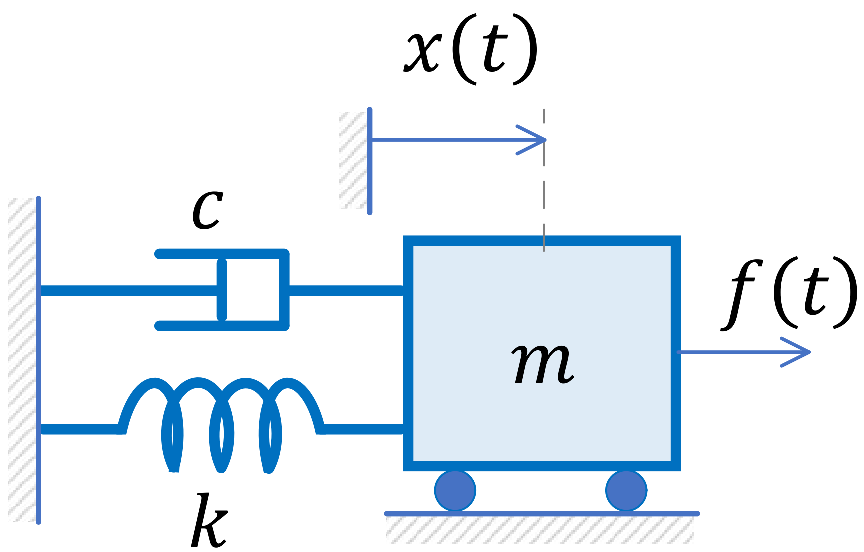
$$f(t) = \begin{cases} 0, & t < 0 \\ \frac{A}{T}t, & 0 \le t < T \\ \frac{A}{2}, & t \ge T \end{cases}$$



$$f(t) = \frac{A}{T}t \cdot h(t) - \frac{A}{2}h(t-T) - \frac{A}{T}(t-T) \cdot h(t-T)$$

$$F(s) = \frac{A}{Ts^2}(1 - e^{-sT}) - \frac{A}{2s}e^{-sT}$$

Primer primene $\mathcal{L}$ na model



$$m \cdot \ddot{x}(t) + c \cdot \dot{x}(t) + k \cdot x(t) = f(t)$$

$$\mathcal{L}\{m \cdot \ddot{x}(t) + c \cdot \dot{x}(t) + k \cdot x(t)\} = \mathcal{L}\{f(t)\}$$
$$m \cdot \mathcal{L}\{\ddot{x}(t)\} + c \cdot \mathcal{L}\{\dot{x}(t)\} + k \cdot \mathcal{L}\{x(t)\} = F(s)$$
$$m(s^2X(s) - sx_0 - v_0) + c(sX(s) - x_0) + kX(s) = F(s)$$
$$(ms^2 + cs + k)X(s) = mx_0s + mv_0 + cx_0 + F(s)$$

$$X(s) = \frac{mx_0s + mv_0 + cx_0 + F(s)}{ms^2 + cs + k}$$

Ovo se svodi na količnik dva polinoma po s.
Konačno, odziv $x(t)$ se dobija primenom inverzne Laplasove transformacije na $X(s)$

$$x(t) = \mathcal{L}\{X(s)\}$$